LAB#16
OPEN ENDED

Objective:

Create a unique scenario and implement it in MIPS by using all concepts covered across the labs, including I/O, arithmetic, branching, loops, bit manipulation, arrays, searching, sorting, control structures (If-Else, For Loop, Switch), and procedures to keep the program modular.

SAMPLE TASKS:

1. Design a student record management system in MIPS Assembly Language. The program should take multiple student marks as input, store them in an array, and use loops to traverse the data. It should perform arithmetic operations to calculate total and average marks, use If-Else conditions to display pass or fail status, apply bit manipulation to set or clear result flags, sort the marks, and search for a specific value entered by the user. A switch structure should allow the user to choose different operations, and procedures should be used to organize tasks such as calculation and display.

2. Develop a sensor monitoring system in MIPS Assembly Language. The program should read multiple sensor readings, store them in an array, and use loops to analyze the data. Arithmetic operations should be used to compute totals or averages, while If-Else conditions determine normal or critical readings. Bit manipulation should be applied to set status flags, and sorting should arrange readings in order. A searching operation should locate specific values, and a switch structure should allow the user to select different analysis modes. Procedures should be used to handle repeated tasks efficiently.

3. Create a menu-driven calculator and data analyzer in MIPS. The program should use input/output to read numbers and user choices, perform arithmetic operations based on a switch control structure, and use procedures for each operation. The user should enter several values that are stored in an array, processed using loops, sorted in a specific order, and searched for a given number. Conditional branching should be used to display appropriate messages, while bitwise operations should be applied to manipulate control flags or values during processing.

Note:

Please give a short explanation of your scenario. Mention what your task does and the main concepts or operations you used.